

Golden Setup: XAUUSD Bot Architecture & Deployment

1. THE BOT ARCHITECTURE (How It Works)

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This algorithm is a mathematically optimized Institutional Momentum trader for XAUUSD (Gold). It consists of three intertwined engines:

A. The Deterministic Engine (ICT / Smart Money Concepts)

The bot tracks the 50/200 Exponential Moving Average to establish the daily bias.

It hunts for "Fair Value Gaps" (FVG) and structural liquidity sweeps exclusively in the direction of the bias.

The Golden Parameters achieved an 83.4% Win Rate on MT5 over 15 months:

- Momentum Filter: ADX MIN = 22
- Quality Score: 40/100 Minimum
- Trailing Stop: Hit 0.5R, lock Break Even, trail runner to 3.5R.

B. The Macro Guardian (ForexFactory News API)

Retail bots lose because they trade during Federal Reserve announcements.

Your bot actively downloads the free ForexFactory XML calendar every hour.

It scans specifically for High Impact (Red Folder) USD data releases (like CPI, NFP, Jerome Powell speeches).

It enforces a total freeze on trading precisely 30 minutes before and after the event.

C. The AI / Machine Learning Engine (HistGradientBoostingClassifier)

Every time the Deterministic Engine finds a trade, the AI Guardian steps in.

The Guardian natively calculates 11 live technical indicators (RSI, multi-EMA divergence, volatility) and reads the live performance of the US Dollar Index (DXY) and 10-Year Bond Yields (^TNX).

It feeds this matrix into the Scikit-Learn .joblib AI models we trained on 4,400+ historical records.

If the AI predicts a Win Rate of less than 65%, the trade is silently aborted.

2. TRANSFERRING TO ANOTHER MAC OR PC (Deployment Steps)

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To move this absolute unit of an algorithm to a high-speed VPS or another computer:

STEP 1: Move the Folder

Copy the entire "XAUUSD_Bot_Project" folder to the Desktop of the new PC/Mac.

STEP 2: Install Python 3

Download and install Python 3.10+ on the new computer.

STEP 3: Configure the Terminal

Open Terminal (Mac) or Command Prompt (Windows). Change directory into the folder:

> cd Desktop/XAUUSD_Bot_Project

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STEP 4: Create the Virtual Environment

You must isolate the AI libraries from the global system.

```
> python3 -m venv .venv
> source .venv/bin/activate      (For Mac)
> .venv\Scripts\activate        (For Windows)
```

STEP 5: Install the AI Dependencies

Install the exact requirements used to train the ML models:

```
> pip install pandas numpy requests scikit-learn joblib flask yfinance
```

STEP 6: One-Click Start

```
> .venv/bin/python bot_free.py
```

(Alternatively on Mac, you can simply run: `bash start_bots.sh`)

STEP 7: Load MT5

Open MetaTrader 5 on the new PC. Open the XAUUSD M15 chart.

Attach the "XAU_Bot" Expert Advisor. Ensure the Algo Trading button is green.

It will instantly connect via TCP to Python Port 9999 and the AI Guardian will take control.